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Modelling of flexible metal wheels for planetary rover on deformable terrain

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ABSTRACT :

Rovers are one of the most important vehicles used for conducting planetary exploration missions. On deformable terrain, traction is limited by the mechanical properties of the soil; therefore, a lack of traction and wheel slippage causes difficulties during operation of the rover. One possible solution for increasing the traction force is to increase the size of the wheel-terrain contact area. Flexible wheel forms can change depending on terrain conditions. So, flexible wheels exhibit high performance in traversing deformable terrain. However, the flexible metal wheel-terrain model for a planetary rover on deformable terrain has not yet been presented in detail. In this paper, a flexible metal wheel and a flexible wheel model that can quantitatively evaluate the sinkage and traction performance of flexible metal wheels on deformable terrain are designed. The theory has been built over the existing equations and validated with by test results. This paper establishes models for estimating static sinkage and dynamic wheel-terrain interaction mechanics. Experimental results from a single-wheel test show that the model can be used for mobility prediction with good accuracy. Based on the calculated model values and experimental results, the drawbar pull and torque of the flexible wheel clearly increases and sinkage clearly decreases compared with a rigid wheel of the same dimensions. The new model can be used to predict the traction performance of flexible wheels. This study can provide a reference for the flexible wheel design of planetary rovers.

1. Introduction

Rovers are one of the most important vehicles used for conducting planetary exploration missions. The tractive force caused by the interaction of the wheel and terrain determines the ability of the planetary rover to climb slopes, accelerate and traverse obstacles. The majority of semi-empirical models use empirical formulations developed by Bekker and Wong [1–6]. Meirion-Griffith et al. presented an improved model of rigid wheels considering the effect of wheel radius on the normal stress [7]. Irani et al. investigated the fluctuation of wheel-terrain interaction caused by lugs and deduced a dynamic model for lightweight vehicle wheels on sandy terrain [8]. Ding et al. proposed a comprehensive model reflecting slip sinkage [9,10].

Based on the analytical rigid wheel models, numerous of flexible wheel theories have been presented. In deformable terrain, traction is limited by the mechanical properties of the soil; therefore, a lack of traction and wheel slippage causes difficulties in the operation of the rover. Flexible wheels improve performance and prevent the planetary rover from becoming stuck in deformable terrain [11]. Flexible wheels can be changeable depending on terrain conditions. So, flexible wheels have high performance in traversing deformable terrain. However, the flexible wheel-terrain model for planetary rover on deformable terrain has not been researched in detail, especially for the flexible metal wheel-terrain model.

At present, some researchers have studied the wheel-soil interaction model of a flexible wheel. Wong et al. built a flexible tyre-terrain interaction model and divided models into a wheel front section, wheel flat section, and wheel rear section, which were calculated separately [12]. The traction force of the wheel is obtained using an integral of the normal and shear stresses generated at each section [12,13]. Muro et al. performed simulations for the interaction between tyres and deformable terrain [14]. Ishigami et al. presented a flexible metal wheel-terrain interaction model and divided the model into three contact sections: front section, flat section, and rear section. Simulation studies with varied wheel pressures were conducted to validate the proposed model [15]. The interaction model of the flexible wheel and deformable terrain was established based on Bekker's equations [2,4].

Currently, some researchers have studied the flexible wheels of the planetary rover. Favaedi et al. have developed of an analytical model which is an extension to the Bekker theory to predict the tractive

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Nomenclature

a	Ellipse semi-major axes *(m)*
b	Ellipse semi-minor axes *(m)*
L	Ellipse radius at angle θ *(m)*
F_N	Normal force of terrain acting on a wheel *(N)*
F_{DP}	Drawbar pull *(N)*
T	Wheel torque *(N·m)*
Z	Wheel sinkage *(m)*
W	Vertical load *(N)*
B	Wheel width *(m)*
n	Sinkage exponent of terrain (dimensionless)
k_c	Cohesive modulus *(N/cm^{n+1})*
k_φ	Friction modulus *(N/cm^{n+2})*
K	Shear deformation modulus *(m)*
φ	Internal friction angle *(°)*
c	Terrain cohesion *(Pa)*
j	Terrain shear displacement *(m)*
s	Slip ratio (dimensionless)
X	A angle formed between horizontal direction and a direction tangential to the ellipse curve *(Rad)*
θ_1	Entrance angle *(Rad)*
θ_2	Exiting angle *(Rad)*
θ_m	Maximum stress angle *(Rad)*
σ	Normal stress of flexible wheel *(Pa)*
σ_1	Normal stress between θ_m and θ_1
σ_2	Normal stress between θ_m and θ_2
σ_m	Maximum normal stress of flexible wheel *(Pa)*
τ	Shear stress of flexible wheel *(Pa)*
τ_1	Shear stress between θ_m and θ_1
τ_2	Shear stress between θ_m and θ_2
τ_m	Maximum shear stress of flexible wheel *(Pa)*

performance for a metal flexible wheel by using the geometric model of the wheel in deformation [13]. The analytical model is based on the ground pressure distribution over three contact sections with the ground. Some experiments have been carried out to simulate actual running conditions using flexible wheels by Kojiro et al. [16]. The experiments were carried out using various flexible wheels that allowed different amounts of deflection. The results showed that flexible wheels had a high level of performance when it was traversing loose soil on a slope. At the same time, Kojiro et al. carried out a performance evaluation for the flexible wheel [16]. Running experiments are conducted to confirm the running performance using an experimental device to evaluate a single wheel. There are many scholars devoted to the design of flexible metal wheels, such as the ESA flexible wheel, DLR flexible wheel, and McGill University flexible wheel. Under a load, a deformable flexible metal wheel is different from a rigid wheel and tyre, but there is little research on the flexible metal wheel-terrain model.

Based on the theories of terramechanics and soil mechanics, we analyse the mechanical characteristics of the interaction between a flexible metal wheel and deformable terrain, and establish a wheel-terrain interaction mechanics model. The analysis of sinkage, drawbar pull, torque and the overall performance of a flexible wheel are considered. Section 2 designs a flexible metal wheel and presents a model to estimate static sinkage when a wheel is subjected to vertical load without torque. Section 3 presents a wheel-terrain interaction mechanics model and estimates dynamic flexible wheel vertical force, drawbar pull and angular torque. The accuracy of the wheel-terrain interaction model is verified. Finally, Section 4 presents the conclusion and summary.

2. Static sinkage estimation of flexible wheel subjected to vertical load without torque

2.1. Wheel-terrain interaction

The interaction between a wheel and deformable terrain is categorized into three situations, as shown in Fig. 1. These categories are distinguished by structural stiffness of the wheel and by the soil stiffness of the terrain. Rigid wheels have a constant diameter or deflections less than 10% of the diameter. A rigid wheel running on deformable terrain is shown in Fig. 1a. Flexible tyre wheels deflect more than 10%, and can be made of non-metallic materials to provide low-to medium-pressure pneumatic tyres [17]. A tyre running on deformable terrain is shown in Fig. 1b. The tyre deflects, and the middle section of the contact patch is assumed to be flat, while the front and rear sections are assumed to maintain a circular shape [18]. A flexible metal wheel running on deformable terrain is shown in Fig. 1c. The flexible metal wheel deflects greater than 10%, and can be made of thin-wall metals to provide the necessary elasticity properties. The interaction between a flexible metal wheel and deformable terrain is approximated by an ellipse.

2.2. Flexible metal wheel design

Due to the peculiar environment on planetary surfaces that consists of strong radiation, large variations in temperature and high vacuum conditions, the wheels of planetary rovers are usually made of metal. As shown in Fig. 2, the flexible wheel considered in this study is based on an all-metallic configuration with three rows of loop springs installed on a rigid hub. The flexible spokes and wheel rim have a damping effect

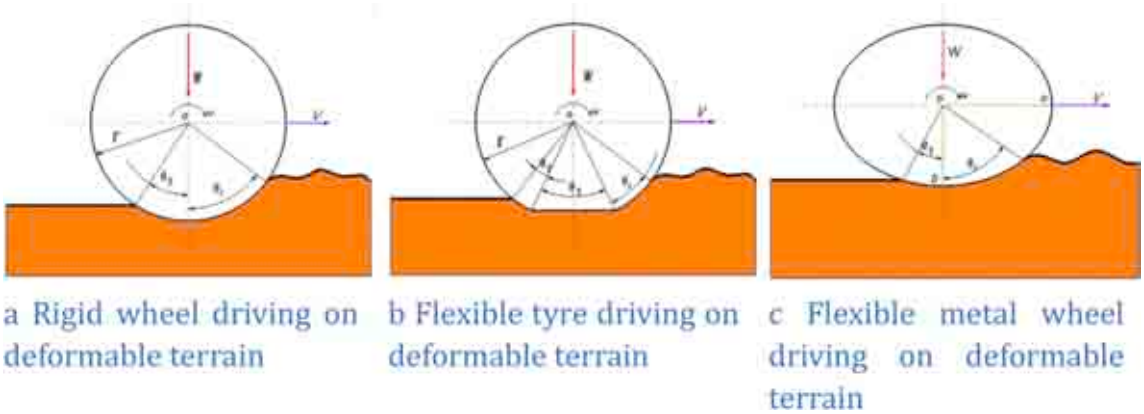

Fig. 1. Wheel-terrain interaction categories.

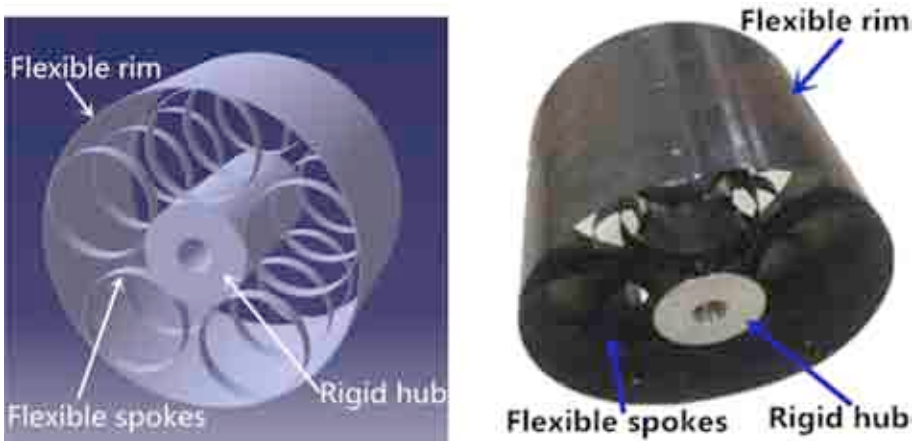

Fig. 2. Flexible wheel model and experimental model.

on the rover, and allow it to smoothly travel on the rough terrain of planets. The wheel width and diameter for the flexible metal wheel are 200 mm and 280 mm, respectively. One end of the three-row spring is connected by bolts to the outer edge of wheel rim, and the other end is bolted to the wheel hub. The flexible wheel design characteristics are shown in Table 1. Flexible metal wheel form can be changeable depending on terrain conditions. To improve wheel elasticity, spring steel is used for the flexible wheel rim and spokes. The mechanical and physical properties of the flexible wheel materials are shown in Table 2. When a load is applied on the flexible metal wheel, its form changes and the part of the wheel in contact with the terrain approximates to an ellipse. The wheels are in a state of outspread augment contact area between the flexible wheel and deformable terrain; therefore, the mobility capability is improved.

2.3. Wheel geometry changes of wheel subjected to vertical load without torque

When the wheel is subjected to a vertical load without torque, the deformation of the bottom of the flexible metal wheel is approximately elliptical, which is different from the traditional tyre deformation, as shown in Fig. 3. The new model approximates the geometrical deformation of the flexible wheel as an ellipse. According to the ellipse geometry relation, the ellipse radius at any point is calculated. Eq. (1) shows the simplified result of the flexible wheel [14].

$$L(\theta) = \frac{ab}{\sqrt{b^2 \sin^2 \theta + a^2 \cos^2 \theta}} \tag{1}$$

where a is the ellipse semi-major axes, b is the ellipse semi-minor axes, θ is angle, and L is the radius of the ellipse at angle θ. As shown in Fig. 3, based on the geometry relation, X can be expressed as

$$X = \arctan\left(\frac{b^2 \tan \theta}{a^2}\right) \tag{2}$$

where X is the angle formed between the horizontal direction and a direction tangential to the ellipse curve.

As the vertical load changes, the semi-major axes a and semi-minor axes b constantly change. The change in the flexible wheel vertical load determines the change in the elliptic semi-major axes a and the semi-minor axes b. Therefore, the semi-major axes a and the semi-minor axes b are related to the vertical load. Experiments were performed using the pressure-sinkage device shown in Fig. 4. The semi-major axes a and the semi-minor axes b of the flexible wheel rim under different loads were measured. The value of a and b is measured with vernier caliper, the range and the accuracy of the vernier caliper is 0–500 mm and 0.01 mm respectively. As shown in Fig. 3, when the vertical load is imposed on the centre of the flexible metal wheel, the flexible wheel becomes increasingly deformed as the load increases, and the shape of the flexible metal wheel becomes an ellipse.

For flexible metal wheels that conform to the integral deformation of the ellipse, the linear relationship between the load and the deformation is shown in Fig. 5, the deformation amount of the flexible wheel has a linear relation with the load of the wheel. The A value of the metal flexible wheels with different parameters is different.

$$W = Ax \tag{3}$$

Eq. (4) shows the deformation amount of the flexible wheel has a linear relation with the load of the wheel. The R-squares of the fitting equation is 0.9756. Eq. (4) can accurately describe the relationship between load and deformation of the wheel.

$$W = 3160.4x \tag{4}$$

where A is the coefficient of deformation, x is the deformation amount of the flexible wheel. According to the geometric relation, the semi-minor axes of elliptic can be expressed as follows

$$b = r - \frac{W}{A} \tag{5}$$

Since the perimeter of the ellipse is constant L, the semi-major axes of elliptic is represented by the ellipse perimeter theorem as follows

$$a = \frac{4b + L - 2\pi b}{4} \tag{6}$$

2.4. Model for estimating static sinkage of wheel subjected to vertical load without torque

When the flexible wheel is subjected to a vertical load without torque, the normal stress under the wheel is symmetrically distributed, and the maximum normal stress occurs at the bottom-dead-centre. As

Table 1
Flexible wheel characteristics.

Design property	Value
Wheel diameter *(mm)*	280
Wheel width *(mm)*	200
Hub diameter *(mm)*	90
Hub width *(mm)*	200
Rim thickness *(mm)*	0.5
Spoke thickness *(mm)*	0.3
Spoke diameter *(mm)*	95
Spoke width *(mm)*	23
Number of spokes	15
Wheel overall mass *(kg)*	3.076
Maximum load design of wheel *(N)*	250

Table 2
Mechanical and physical properties of flexible wheel materials.

Wheel parts	Material	Density/$g{\cdot}cm^{-3}$	Yield strength/*MPa*	Tensile strength/*MPa*	Elongation rate *(%)*	Reduction of area
Hub	Aluminium 2024	*2.73*	*275*	*425*	*15*	-
Rim	Spring steel 65Mn	*7.85*	*785*	*980*	*8*	*30*
Spoke	Spring steel 65Mn	*7.85*	*785*	*980*	*8*	*30*

shown in Fig. 6, $\theta_m = 0$, $\theta_1 = -\theta_2$, and the shear stress is small due to no rotation. The horizontal force acting on the wheel becomes zero because the horizontal components of the normal and shear stresses are symmetrically cancelled out. The vertical load is the integration of the normal stress, and Eq. (7) becomes

$$W = 2b\int_0^{\theta_1} \sigma_1 L(\theta)\cos X d\theta \tag{7}$$

According to the ellipse geometry relation, the wheel sinkage is calculated.

$$Z = b - L_1\cos\theta_1 = b - \frac{ab\cos\theta_1}{\sqrt{b^2\sin^2\theta_1 + a^2\cos^2\theta_1}} \tag{8}$$

The maximum normal stress σ_m can be expressed as

$$\sigma_m = \left(\frac{k_c}{B} + k_\phi\right)Z^n = \left(\frac{k_c}{B} + k_\phi\right)\left(b - \frac{ab\cos\theta_1}{\sqrt{b^2\sin^2\theta_1 + a^2\cos^2\theta_1}}\right)^n \tag{9}$$

To calculate the integral model, the normal stress can be simplified [19–21]. The simplified normal stress can be expressed as

$$\begin{cases}\sigma_1(\theta) = \sigma_m(\theta_1 - \theta)/(\theta_1 - \theta_m) & (\theta_m \le \theta \le \theta_1)\\ \sigma_2(\theta) = \sigma_m(\theta - \theta_2)/(\theta_m - \theta_2) & (\theta_2 \le \theta \le \theta_m)\end{cases} \tag{10}$$

Substituting Eq. (9) and Eq. (10) into Eq. (7) yields an expression for the vertical load, written as Eq. (11)

$$W = 2B\left(\frac{k_c}{B} + k_\varphi\right)\cdot\left(b - \frac{ab\cos\theta_1}{\sqrt{b^2\sin^2\theta_1 + a^2\cos^2\theta_1}}\right)^n \int_0^{\theta_1} \frac{\theta_1 - \theta}{\theta_1}\cdot\cos X \cdot\frac{ab}{\sqrt{b^2\sin^2\theta + a^2\cos^2\theta}}d\theta \tag{11}$$

2.5. Pressure–sinkage test

To evaluate the static sinkage model, a flexible pressure-sinkage test was carried out, as shown in Fig. 4. Simultaneously, a contrasting experiment of the rigid wheel with the same size was added. Table 3 shows the parameters of the test soil.

The experimental parameters of the flexible wheel vertical load and load speed are shown in Table 4.

To evaluate the model, it is necessary to calculate the sinkage depth

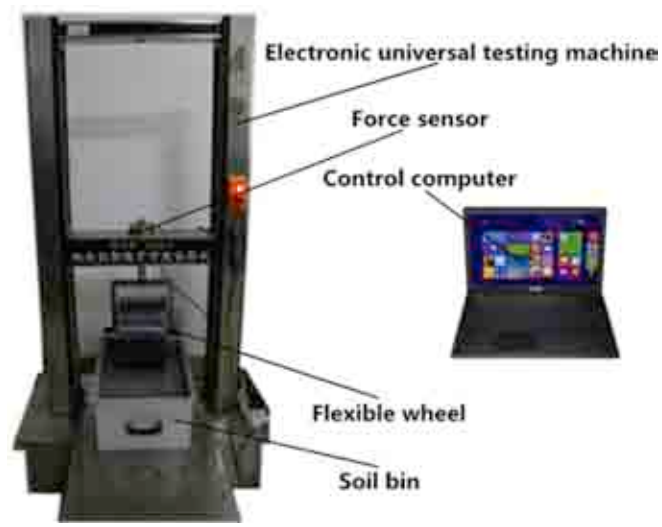

Fig. 4. Pressure-sinkage testbed.

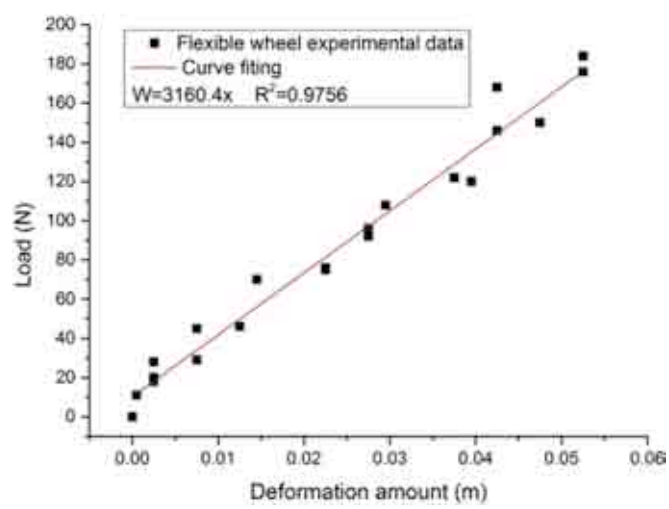

Fig. 5. Relationship between vertical wheel load and elliptic deformation.

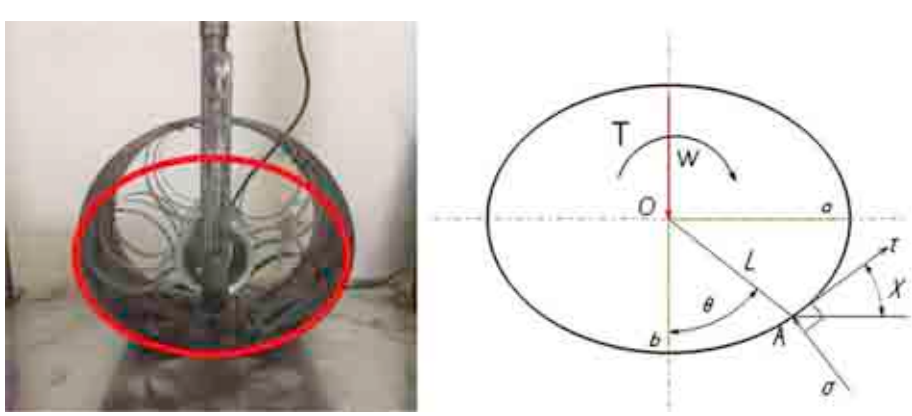

Fig. 3. Wheel rim ellipse.

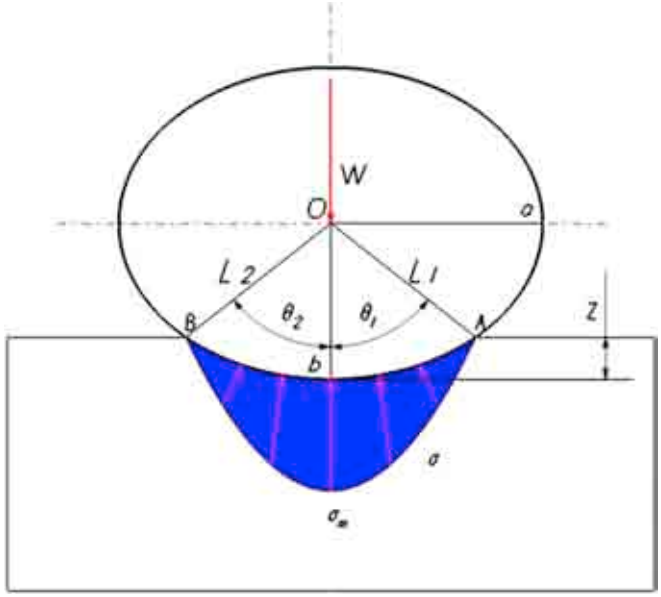

Fig. 6. Stress distribution for flexible wheel on vertical load.

Table 3
Parameters of test soil.

Parameter	Value
Bulk density *(g·cm^{-3})*	*1.26*
n (dimensionless)	*1.04*
c (KPa)	*2.14*
k_φ *(N/cm^{n+2})*	*0.84*
k_c *(N/cm^{n+1})*	*1.64*
φ (°)	*28.6*
K (cm)	*1.806*

Table 4
Experimental parameters.

Test parameter	Value
Vertical load *(N)*	*20,40,60,80,100,120,140,160*
Load speed *(mm/min)*	*10*
Repeat test	*5*

of the wheel rut under different vertical loads. The rut morphology under different vertical loads was scanned using a handheld laser scanner (Curise 3D, USA), whose accuracy the accuracy is 0.02 mm , as shown in Fig. 7a. Fig. 7b is the wheel-tracking sinkage obtained using the software Image Process after image digitization. The test was

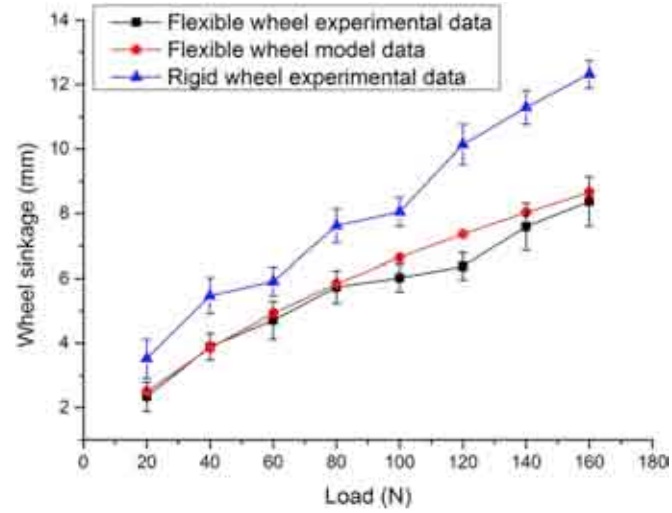

Fig. 8. Wheel sinkage under different vertical loads.

repeated five times per group and 8 groups of wheel sinkage data was collected for the flexible wheel under different vertical loads.

2.6. Model calculation procedure

First, the vertical load, test soil parameters and wheel parameters of the flexible wheel are input to Eq. (8). Then, the entrance angle θ_1 is calculated using Matlab, and the model sinkage data is obtained by inputting θ_1 to Eq. (5). The experimental data and calculated model value of the flexible wheel sinkage are shown in Fig. 8. As shown in Fig. 8, the model calculations for the sinking of flexible wheels under different vertical loads are obtained.

2.7. Static sinkage analysis of wheel subjected to vertical load without torque

Fig. 8 shows sinkage increase as the vertical load increase. Wheel sinkage for a rigid wheel is higher than for a flexible wheel under the same vertical load. When the vertical load is 20 N, the flexible wheel experimental sinkage is approximately 2.3 mm according to the model data, but less than the rigid wheel experimental sinkage of 3.5 mm. When the vertical load is greater than 80 N, the difference between the experimental data and model calculations for the sinkage of the flexible wheel increases. When the vertical load is 160 N, the wheel sinkage for the flexible wheel is 8.66 mm from the experiment and 8.37 mm from the model, the sinkage for a rigid wheel nearly reached 12.3 mm in the experiment. Elastic deformation under vertical load can increase the ground area, reducing grounding pressure and reducing sinkage. It can be seen from the model calculations and experimental results that the

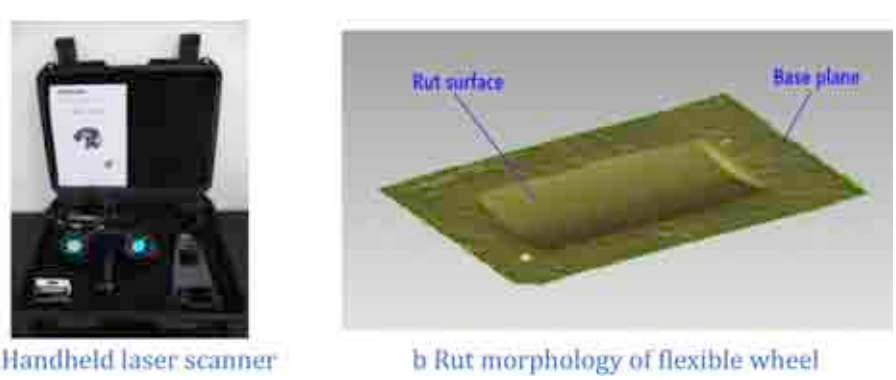

Fig. 7. Handheld laser scanner and rut morphology of flexible wheel.

sinkage of the flexible wheel was reduced by 31.45% compared with that of a rigid wheel of the same dimensions.

Fig. 8 shows that the mean relative error (MRE) of the flexible wheel model is 6.30% and that the root mean squared error (RMSE) of the flexible model is 0.67 mm. Based on this analysis, the static sinkage model of the flexible wheel conforms to the pressure-sinkage data with good accuracy and has relatively lower mean relative error and root mean squared error values. This model can quantitatively evaluate the sinkage of a flexible metal wheel on deformable terrain.

3. Wheel-terrain interaction model of flexible wheel

When the flexible wheel is in motion, the maximum stress position is in the front of the bottom-dead-centre, due to the wheel slip. According to the normal stress and shear stress distribution of the contact stress distribution on the flexible wheel, the vertical load model, torque model and drawbar pull model of the flexible wheel are obtained using the numerical integration method for the entire contact patch. Therefore, the problem of using series approximation in the integration of the Bekker model is avoided [18].

3.1. Establishing wheel-terrain interaction model

For simplicity, the flexible wheel shape is assumed to be an ellipse. As shown in Fig. 9, given a constant rover forward speed and vertical load, based on the balance of force and moment, the interaction model of flexible wheel traversing deformable terrain is obtained.

Fig. 9 shows the flexible wheel travel through deformable terrain, where Z is sinkage, a is the ellipse semi-major axes and b is the ellipse semi-minor axes. Based on the force and moment balance, the interaction integral model of flexible metal wheels for a planetary rover on deformable terrain is given.

$$W = B\int_0^{\theta_1} L(\sigma_1(\theta)\cos X + \tau_1(\theta)\sin X)d\theta + B\int_0^{\theta_2} L(\sigma_2(\theta)\cos X - \tau_2(\theta)\sin X)d\theta \tag{12}$$

$$F_{DP} = B\int_0^{\theta_1} L(\tau_1(\theta)\cos X - \sigma_1(\theta)\sin X)d\theta + B\int_0^{\theta_2} L(\tau_2(\theta)\cos X + \sigma_2(\theta)\sin X)d\theta \tag{13}$$

$$T = B\int_0^{\theta_1} L^2\tau_1(\theta)d\theta + B\int_0^{\theta_2} L^2\tau_2(\theta)d\theta \tag{14}$$

3.2. Simplified analytic model

The shear stress and normal stress change with the exiting angle and entrance angle when driving on deformable terrain. The entrance angle can be expressed as

$$\theta_1 = \arctan\sqrt{\frac{a^2}{(b-Z)^2} - \frac{a^2}{b^2}} \tag{15}$$

Iagnemma et al. presented a method for calculating the integrals by simplifying the normal stress and shear stress [18,20,22,23]. Based on this observation, the simplified normal stress and shear stress equations can be expressed as [24]

$$\begin{cases} \sigma_1(\theta) = \sigma_m(\theta_1 - \theta)/(\theta_1 - \theta_m) & (\theta_m \le \theta \le \theta_1) \\ \sigma_2(\theta) = \sigma_m(\theta - \theta_2)/(\theta_m - \theta_2) & (\theta_2 \le \theta \le \theta_m) \\ \tau_1(\theta) = \tau_m(\theta_1 - \theta)/(\theta_1 - \theta_m) & (\theta_m \le \theta \le \theta_1) \\ \tau_2(\theta) = \tau_m(\theta - \theta_2)/(\theta_m - \theta_2) & (\theta_2 \le \theta \le \theta_m) \end{cases} \tag{16}$$

The location of the maximum normal stress can be expressed as

$$\theta_m = \frac{\theta_1 + \theta_2}{2} \tag{17}$$

The maximum normal stress of the flexible wheel can be expressed as

$$\sigma_m = \left(\frac{k_c}{B} + k_\varphi\right)\left(\frac{ab\cos\frac{\theta_1}{2}}{\sqrt{b^2\sin^2\frac{\theta_1}{2} + a^2\cos^2\frac{\theta_1}{2}}} - \frac{ab\cos\theta_1}{\sqrt{b^2\sin^2\theta_1 + a^2\cos^2\theta_1}}\right)^n \tag{18}$$

In this paper, the exiting angle of the wheel θ_2 is assumed to be 0, since θ_2 is very small when the flexible wheel is driving on low cohesive terrain. The shear stress at an arbitrary contact point of the wheel and terrain can be expressed by positive stress [25].

$$\tau = (c + \sigma\tan\varphi)\left(1 - e^{-\frac{j}{K}}\right) \tag{19}$$

According to the shear stress distribution of the terrain under the flexible wheel, using the relationship between the flexible wheel slip ratio and shear displacement, shear displacement can be expressed as [26,27]

$$j = \int_{\frac{\theta_1}{2}}^{\theta_1} L[1 - (1 - s)\cos X]d\theta \tag{20}$$

where s is the slip ratio and j is the terrain shear displacement. Assuming the maximum normal stress and maximum shear stress are at the same point of the flexible wheel rim, another relationship is established.

$$\tau_m = (c + \sigma_m\tan\varphi)\left(1 - e^{-\frac{j}{K}}\right) \tag{21}$$

Fig. 9 shows normal stress and shear stress distribution under the flexible wheel. The maximum normal stress σ_m divides the normal stress in two parts. In part I, normal stress σ_1 occurs between θ_1 and θ_m. In part II, normal stress σ_2 occurs between θ_m and θ_2. Force and moment balance equations can be written for the system shown in Fig. 9. The shear stress and normal stress equations over the appropriate contact areas is integrated. Eqs. (12)–(14) are too complex to be integrated. Substituting Eqs. (15)–(21) into Eqs. (12)–(14). Simplified forms of the vertical load, torque and drawbar pull of the interaction model can be expressed as

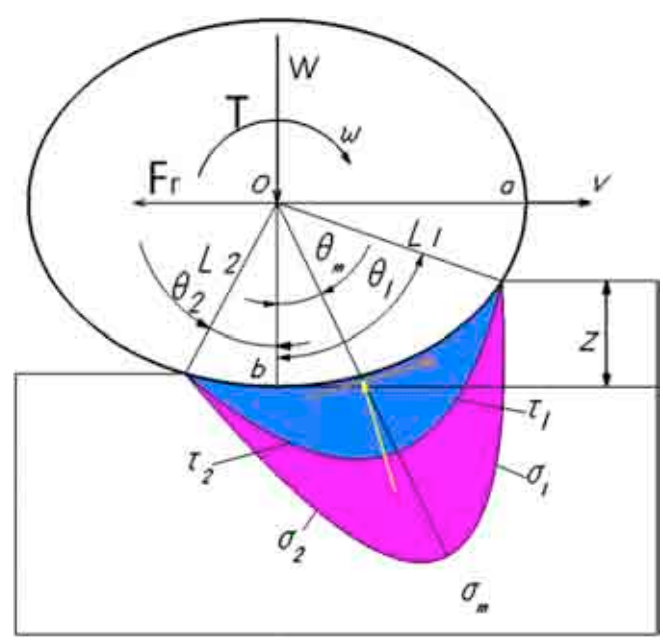

Fig. 9. Analytical model of flexible wheel-terrain interaction.

$$W = B\int_0^{\frac{\theta_1}{2}}\left(\frac{2\sigma_m\cdot\theta\cdot L\cdot\cos X}{\theta_1} + \frac{2\tau_m\cdot\theta\cdot L\cdot\sin X}{\theta_1}\right)d\theta + B\int_{\frac{\theta_1}{2}}^{\theta_1}\left(\frac{2\sigma_m\cdot(\theta_1-\theta)\cdot L\cdot\cos X}{\theta_1} + \frac{2\tau_m\cdot(\theta_1-\theta)\cdot L\cdot\sin X}{\theta_1}\right)d\theta \tag{22}$$

$$F_{DP} = B\int_0^{\frac{\theta_1}{2}}\left(\frac{2\tau_m\cdot\theta\cdot L\cdot\cos X}{\theta_1} - \frac{2\sigma_m\cdot\theta\cdot L\cdot\sin X}{\theta_1}\right)d\theta + B\int_{\frac{\theta_1}{2}}^{\theta_1}\left(\frac{2\tau_m\cdot(\theta_1-\theta)\cdot L\cdot\cos X}{\theta_1} - \frac{2\sigma_m\cdot(\theta_1-\theta)\cdot L\cdot\sin X}{\theta_1}\right)d\theta \tag{23}$$

$$T = B\int_0^{\frac{\theta_1}{2}}\frac{2\tau_m\cdot\theta\cdot L^2}{\theta_1}d\theta + B\int_{\frac{\theta_1}{2}}^{\theta_1}\frac{2\tau_m\cdot(\theta_1-\theta)\cdot L^2}{\theta_1}d\theta \tag{24}$$

3.3. Wheel-terrain interaction test

The Ministry of Education Key Laboratory of Bionic Engineering of Jilin University developed the single-wheel testbed shown in Fig. 10 [28]. The trolley was connected to the slide guides of the soil bin through linear bearings. The test wheel is driven by a stepper motor directly, and the thrust generated from wheel soil interaction drive the whole trolley forward. The trolley is instrumented with vertical displacement sensor (XPS-S-5000-11R0), horizontal displacement sensor (D38H-8G05L-360BM-5M), rotation sensor (DVC37T), force sensor (HT-7311S1) and torque sensor (CNY-81). The vertical and horizontal displacement sensor is used to measure the sinkage and the horizontal displacement of the wheel, respectively. The trolley adopted a lever structure, the static wheel vertical load can be adjusted by the mass and position of the bob-weight. The vertical load of wheel will be fluctuant when the wheel is running, therefore, vertical load of the dynamic wheel should be measured in real time by the force sensor. The driving torque and the rotation rate of the motor are measured by the torque sensor and rotation sensor, respectively.

Due to the low gravity environment, the vertical load of planetary rover wheel is generally small. Considering load fluctuation due to the different pose positions and orientations of the planetary rover, the vertical load of the flexible wheel is set equal to 40 N, 80 N and 120 N. Table 3 shows the test soil parameters. The soil for testing was made from volcanic rock/ash, derived from Jinchuan city in Jilin province, China. To obtain test soil, the original volcanic rock/ash need to be processed by a series of procedures through a series of processing, including removing impurities, grinding, sieving, drying, mixing and matching. Unmanned planetary rover exploration and route planning depend on the analysis and assessment of image information obtained from the rovers. Therefore, the velocity of current planetary rovers is 25 mm/s. As the travel speed is constant, the slip ratio is controlled by the drawbar pull, which is called traction load method. Depending on the vertical load, the range of experimental slip seems to be varied. The experimental parameters of the flexible wheel speed and vertical load are shown in Table 5.

Table 5
Experimental parameters.

Test parameter	Value		
Vertical load (*N*)	*40*	*80*	*120*
Speed (*mm/s*)	*25*	*25*	*25*
Slip ratio (dimensionless)	*0–0.5*	*0–0.5*	*0–0.35*
Repeat test	*5*		

The test can be used to obtain the flexible wheel torque, drawbar pull, horizontal displacement, rotational speed, slip ratio, sinkage, etc. The total displacement of the driving wheel is 1000 mm. To eliminate the influence of motor initiating and braking, valid data is selected at a displacement range of 200–800 mm. The tests are carried out at least five times to ensure the repeatability of the experiments. The experiments were verified to have good repeatability [28,29]. Fig. 11 shows the actual sinking of the wheel extracted using images. Sinkage Z is based on measuring the height difference between the flexible wheel contact and driving process. The slip ratio of the flexible wheel is calculated using the actual displacement of the wheel and theoretical displacement data.

3.4. Model calculation procedure

First, input load W to Eq. (2) to determine ellipse semi-major axes a and semi-minor axes b. Substitute experimental sinkage Z into Eq. (15) and obtain the entrance angle θ_1. Then, the maximum shear stress τ_m and maximum normal stress σ_m are obtained by substituting the entrance angle θ_1 into Eqs. (16)–(21). Finally, the slip ratio, maximum normal stress σ_m and maximum shear stress τ_m are substituted into Eqs. (22)–(24), and the calculated model values of vertical load, drawbar pull and torque are obtained.

Because Eqs. (22)–(24) are too complicated, Matlab is used to calculate Eqs. (22)–(24). The calculated model values of the vertical load, torque and drawbar pull are obtained.

3.5. Results and analysis

To verify the accuracy of the model, the flexible wheel is tested separately for three vertical loads. To achieve different sliding conditions of the flexible wheel, this study uses the drawbar pull to control the wheel slip ratio. The model calculated value of vertical load, drawbar pull and torque are obtained by substituting the test values of the vertical load, sinkage and slip rate into the model. The model calculated value is compared with the test values of the flexible wheel and rigid wheel.

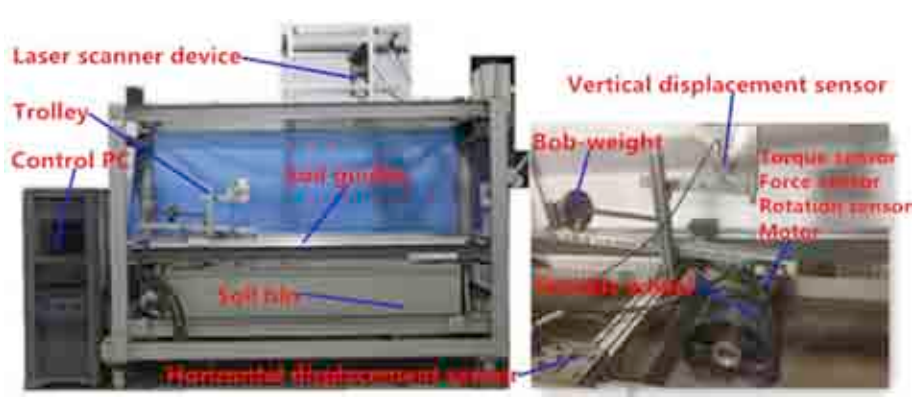

Fig. 10. Wheel-terrain testbed.

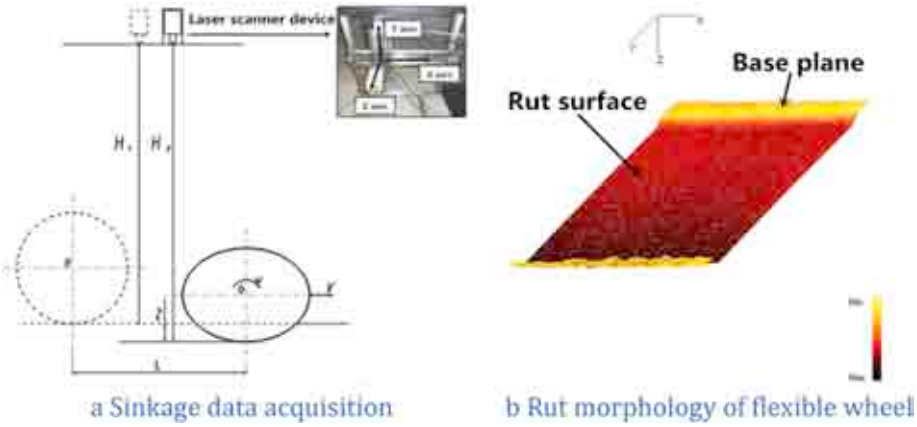

Fig. 11. Sinkage data acquisition and rut morphology of flexible wheel.

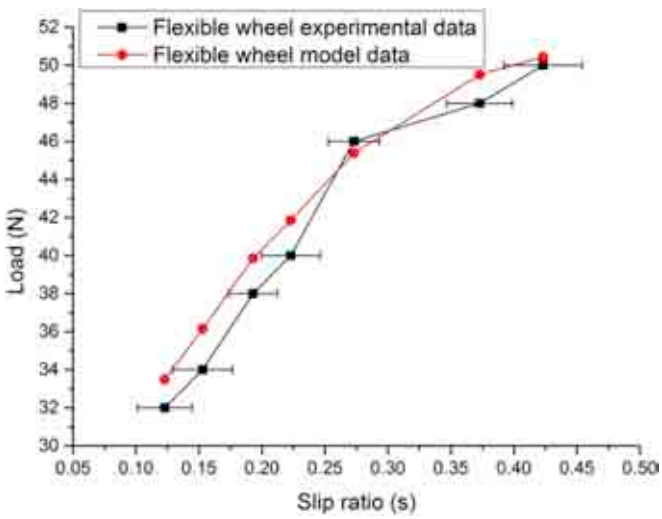

Fig. 12. Relationship between vertical load and slip ratio for a 40 N load.

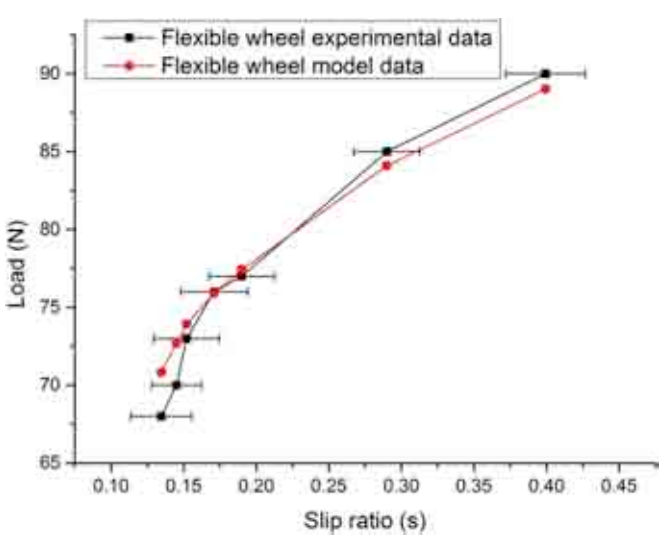

Fig. 13. Relationship between vertical load and slip ratio for an 80 N load.

Figs. 12–14 show the load model calculated value and experimental results for three vertical loads on the flexible wheel on test soil. The model calculated value of load increases as the flexible wheel slip ratio increases. This is because as the slip ratio increases, the flexible wheel sinkage caused by the slip increases, and the contact area between the wheel and soil increases. Thus, the vertical component of soil force acting on the wheel increases, and the model calculated value of vertical load increases with slip ratio. Comparing the model value with the test value, the MREs of the model calculated value with 40 N, 80 N and 120 N loads are 17.61%, 9.513% and 28.27%, respectively. The reason of the difference between model value and test value can be divided into the two aspects, the first one is that error is caused when the model is established, and another one is that error can be also produced when the test is operating.

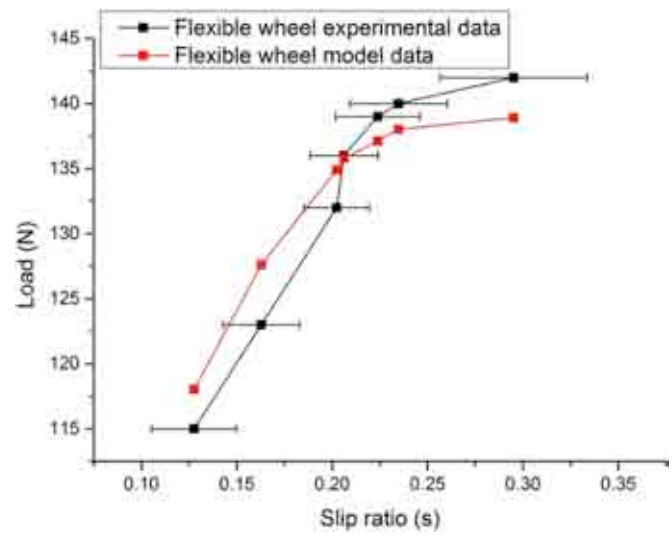

Fig. 14. Relationship between vertical load and slip ratio for a 120 N load.

Figs. 15–17 illustrate the model calculated value and experimental results for the result of drawbar pull on the flexible wheel on the test soil. The model calculated value of drawbar pull increases as the slip ratio increases. This is because as the slip ratio increases, the flexible wheel sinkage caused by the slip increases, and the contact area between the wheel and soil increases. Thus, the horizontal component of the soil force acting on the wheel increases, and the model calculated value of drawbar pull increases with slip ratio. Comparing the model calculated value with the test value, the MREs of the model with 40 N, 80 N and 120 N loads are 23.32%, 13.99% and 17.55%, respectively. The model of drawbar pull can accurately predict the drawbar pull of a flexible metal wheel on deformed terrain. From the experimental results, the drawbar pull of the flexible wheel clearly increases compared with a rigid wheel of the same dimensions.

Figs. 18–20 illustrate the model calculated value and experimental results for three vertical loads on the flexible wheel on test soil. The model calculated value of torque increases as the flexible metal wheel

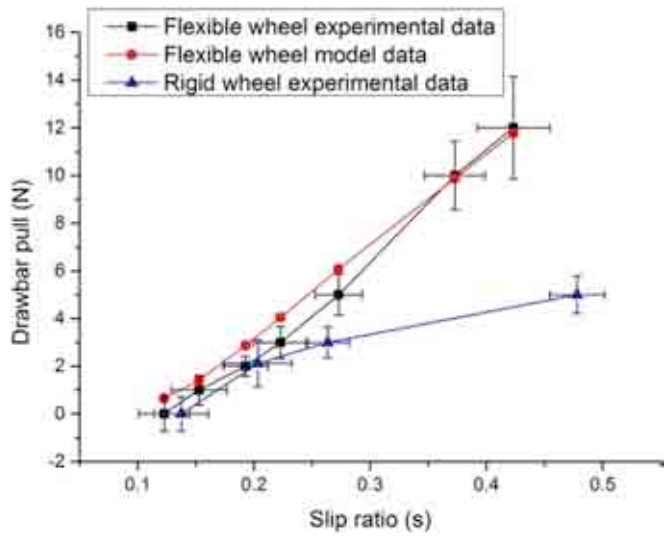

Fig. 15. Relationship between drawbar pull and slip ratio for a 40 N load.

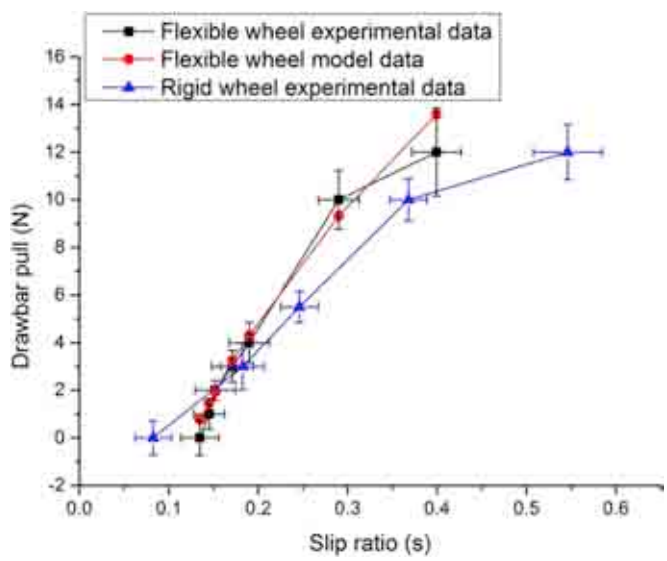

Fig. 16. Relationship between drawbar pull and slip ratio for an 80 N load.

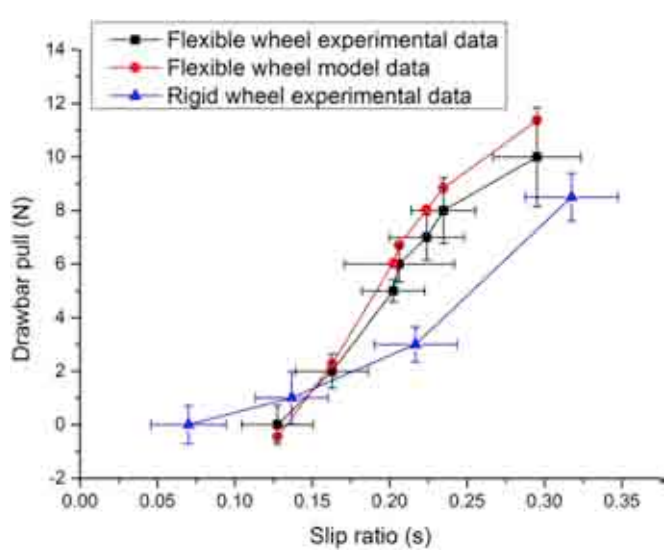

Fig. 17. Relationship between drawbar pull and slip ratio for a 120 N load.

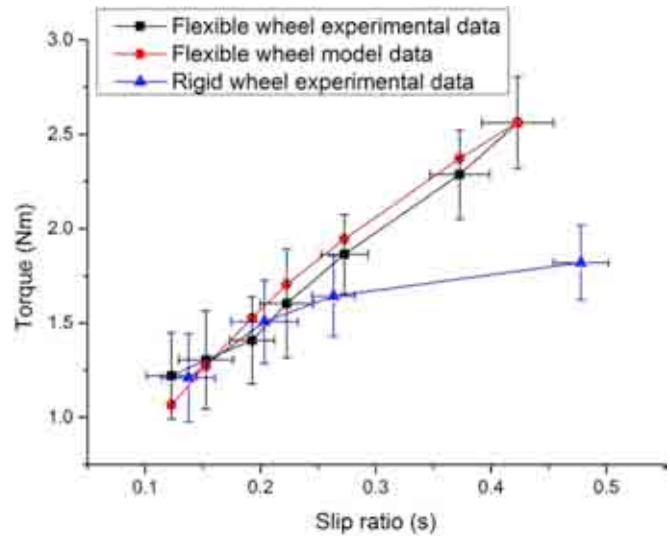

Fig. 18. Torque of simulation and experiment for a 40 N load.

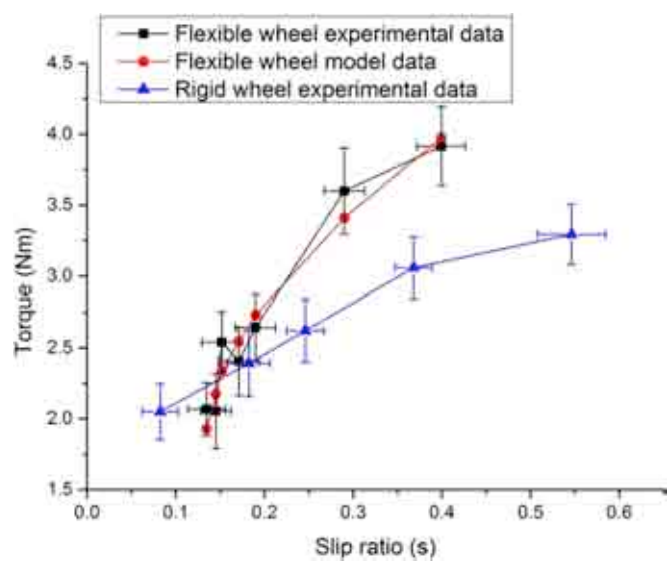

Fig. 19. Torque of simulation and experiment for an 80 N load.

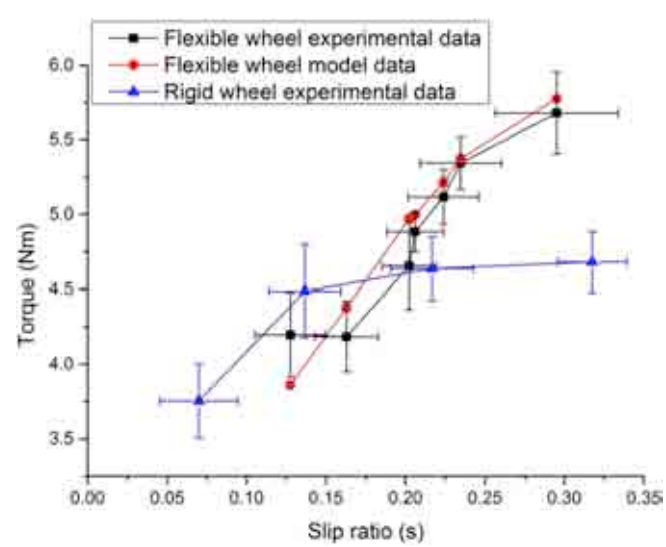

Fig. 20. Torque of simulation and experiment for a 120 N load.

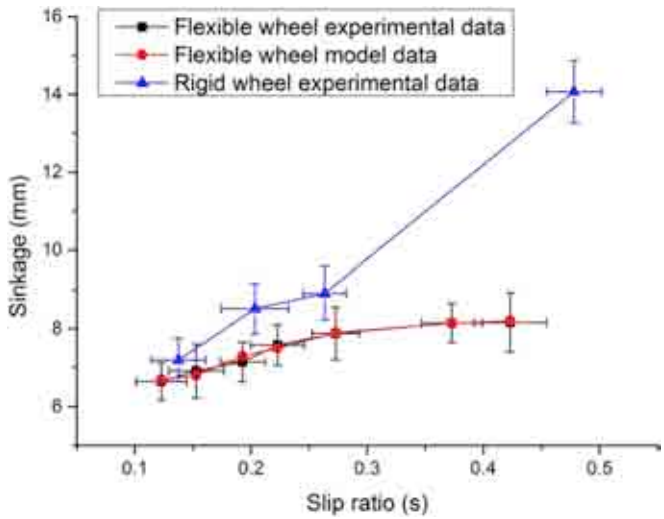

Fig. 21. Sinkage of rigid wheel and flexible metal wheel for a 40 N load.

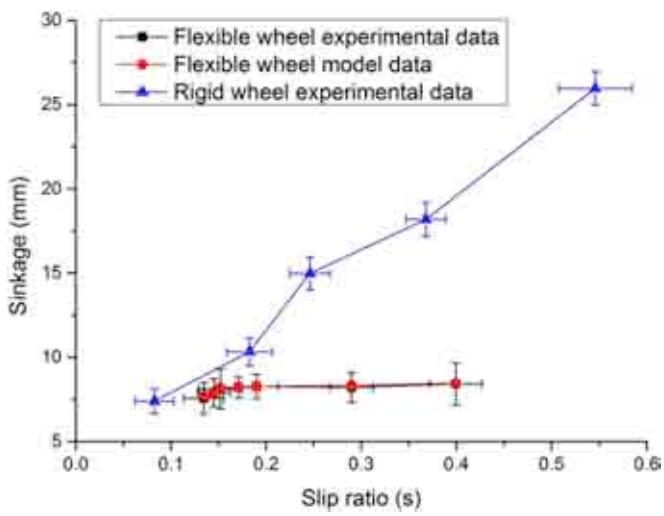

Fig. 22. Sinkage of rigid wheel and flexible metal wheel for an 80 N load.

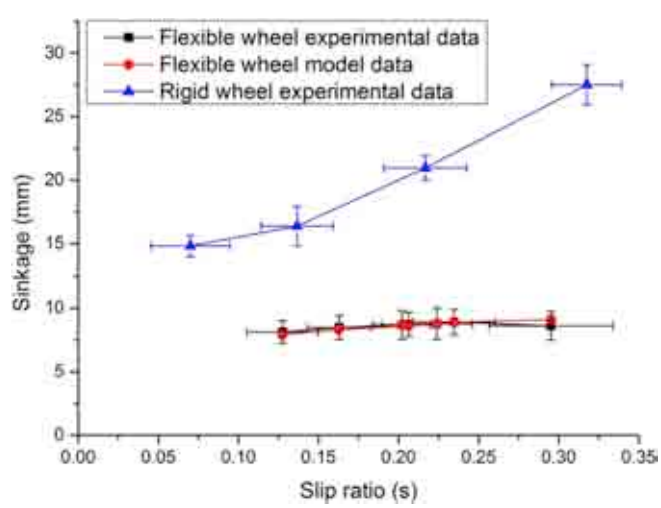

Fig. 23. Sinkage of rigid wheel and flexible metal wheel for a 120 N load.

slip ratio increases. This is because as the slip ratio increases, the flexible wheel sinkage caused by the slip increases, and the contact area between the wheel and soil increases. Thus, the torque acting on the wheel increases, and the model calculated value of torque increases with slip ratio. Comparing the model calculated value with the test value, the MREs of the model with 40 N, 80 N and 120 N load are 10.87% and 6.50%, 4.76%, respectively. Drive torque for a flexible wheel is generally higher than the predicted values because a part of energy is expended in flexing and unflexing of the wheel and gets damped in the process. The model of torque can accurately predict the torque of a flexible metal wheel on deformed terrain. From the experimental results, the torque of the flexible wheel clearly increases compared with a rigid wheel of the same dimensions.

Figs. 21–23 illustrate the flexible wheel and rigid wheel experimental results for sinkage on the flexible wheel on test soil. As the slip ratio increases, the sinkage of the flexible wheel and rigid wheel increases, and the rate of change in rigid wheel sinkage is larger than that of the flexible wheel. Figs. 21–23 show that the mean relative error (MRE) of the flexible wheel model is 5.70% and that the root mean squared error (RMSE) of the flexible model is 0.58 mm. Dynamic sinkage of the flexible wheel is caused by the dynamic movement of the soil under the wheel, depending on the slip ratio and terrain characteristics. From the experimental results, compared with a rigid wheel with the same dimensions, the sinkage of the flexible wheel clearly decreases. The flexible metal wheel form can be changeable depending on terrain conditions. Flexible metal wheels can increase the area of the contact patch between the wheel and terrain. Therefore, flexible metal wheels have good performance for traversing deformable terrain.

4. Conclusions

In this paper, the design of a flexible metal wheel and a new interaction model for flexible metal wheel driving on deformable terrain are presented. The flexible metal wheel form can change depending on terrain conditions. Flexible metal wheels can increase the contact area between the wheel and terrain to reduce the pressure and reduce sinkage. The torque and drawbar pull of the flexible wheel clearly increases compared with a rigid wheel of the same dimensions. So, flexible wheels have high performance for traversing deformable terrain. The new geometric model approximates the flexible wheel rim as an ellipse. In this paper, a model to estimate sinkage when a wheel is subjected to a vertical load without torque is established. When the wheel is subjected to different loads and torques, a large number of dynamic tests are performed. The wheel is deformed as an ellipse when subjected to loads and torques. Based on this, a wheel-terrain interaction model of a flexible wheel is established. To verify the interaction mechanics model, the results of the model calculated value are compared with a single-wheel test. Experimental results from a single-wheel test show the model can be used for mobility prediction with good accuracy. From the model, the calculated value and experimental results show the tractive force of the flexible wheel clearly increases and sinkage clearly decreases compared with a rigid wheel with the same dimensions. Because the flexible metal wheel increases the contact patch area between the wheel and terrain, a stronger traction force is generated. The new model can be used to predict the traction performance and sinkage of flexible wheels and as a reference for design.

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Appendix A. Matlab code for estimating static flexible wheel sinkage

```
clc

clear all

close all
%% parameters
W = 120; B = 20;  n = 1.04;

K_c = 1.64;K_f = 0.840;

lim = 0.1;

syms z

x = 0:0.0001:10;

a = (0.0831 * W + 142.27)/10;   %The ellipse semi-major axes (m)

b = (-0.2766 * W + 140.8)/10;   %The ellipse semi-minor axes (m)

for i = 1:length(x)

    fun = @(z)((x(i)-z)./x(i) .* cos(atan(((b^2)*tan(z))/a^2))...
        .* a.*b./sqrt(power(b,2)*power(sin(z),2)+power(a,2).*power(cos(z),2))/100);

    M =2*B*100*(K_c/B+K_f)*( b-(a*b*cos(x(i))/...
        sqrt(b^2*sin(x(i))^2+a^2*cos(x(i))^2)) )^n * ...

        integral(fun,0,x(i));

    if abs(M - W) < lim

        break;

    end

end

x(i)                            %The entrance angle (Rad)

Z=( b-(a*b*cos(x(i))/sqrt(b^2*sin(x(i))^2+a^2*cos(x(i))^2)) )
```

Appendix B. Matlab code for the wheel-terrain interaction mechanics model

```
clc
clear all
close all
%%   parameters
W =40;                          %Vertical wheel load (N)

z =0.6642308;                   %Wheel sinkage (m)

S =0.122781429;                 %Slip ratio (dimensionless)

B = 20;                         %Wheel width (m)

C =2140;                        %Soil cohesion (Pa)

F = 0.499164;                   %Internal Friction angle (° )

K =1.4;                         %Soil deformation modulus due to sinkage (m)

n = 1.04;                       %Sinkage exponent of terrain (dimensionless)

K_c = 1.64;                     %Cohesive modulus(N/cmn+1)

K_f = 0.840;                          %Friction modulus(N/cmn+2)  %% Fuction
a = (0.0831 * W + 142.27)/10;         %The ellipse semi-major axes (m)

b =(-0.2766 * W + 140.8)/10;          %The ellipse semi-minor axes (m)

theta = atan( sqrt( a^2/(b-z)^2 - a^2/b^2 ) ); %The entrance angle (Rad)

theta_m = 10000*( K_c/B + K_f) * ( a*b*cos(theta/2) / ...
    sqrt( b^2*sin(theta/2)^2 + a^2 *cos(theta/2)^2 ) - ...
    a*b*cos(theta) / sqrt( b^2*sin(theta)^2 + ...
    a^2 * cos(theta)^2 ) )^n;
%Maximum normal stress acting on radial direction of wheel (Pa)
syms x

L = (a.*b) ./ sqrt(power(b,2).*power(sin(x),2)+...
    power(a,2).*power(cos(x),2)); %Radius of the ellipse at angle θ (m)

fun = @(x)((a.*b) ./ sqrt(power(b,2).*power(sin(x),2)+power(a,2)...
    .*power(cos(x),2))).*(1-(1-S).*cos(atan(((b^2)*tan(x))/a^2)) );

J = integral(fun,theta/2,theta);  %Soil shear displacement (m)
```

```
delay_m =( C + theta_m * tan(F) )*(1-exp(-J/K));
%Maximum shear stress acting on radial direction of wheel(Pa)

fun2 = @(x)( 2.*theta_m.*x.*(((a.*b) ./ sqrt(power(b,2) ...
    .*power(sin(x),2) +power(a,2).*power(cos(x),2)))./100)...
    .*cos(atan(((b^2)*tan(x))/a^2))./theta + 2.*delay_m.*x.*(((a.*b) ./ ...
    sqrt(power(b,2).*power(sin(x),2)+power(a,2).*power(cos(x),2)))./100)...
    .*sin(atan(((b^2)*tan(x))/a^2))./theta);
fun3 = @(x)( 2.*theta_m.*(theta-x).*(((a.*b) ./ sqrt(power(b,2)...
    .*power(sin(x),2)+power(a,2).*power(cos(x),2)))./100)...
    .*cos(atan(((b^2)*tan(x))/a^2))./theta+ 2.*delay_m.*(theta-x).*(((a.*b)...
    ./ sqrt(power(b,2).*power(sin(x),2)+power(a,2).*power(cos(x),2)))./100)...
    .*sin(atan(((b^2)*tan(x))/a^2))./theta);

Y = (B/100)*integral(fun2,0,theta/2)+(B/100)*integral(fun3,theta/2,theta);
%Normal force of terrain acting on a wheel (N)

fun4 = @(x)( 2.*delay_m.*x.*(((a.*b) ./ sqrt(power(b,2).*power(sin(x),2)...
    +power(a,2) .*power(cos(x),2)))./100).*(((a.*b) ./ sqrt(power(b,2)...
    .*power(sin(x),2)+power(a,2).*power(cos(x),2))/100)/theta));
fun5 = @(x)( 2.*delay_m.*(theta-x).*((a.*b) ./ sqrt(power(b,2)...
    .*power(sin(x),2)+power(a,2).*power(cos(x),2))./100).*(((a.*b)...
    ./ sqrt(power(b,2).*power(sin(x),2)+power(a,2).*power(cos(x),2))./100)/theta));

I = (B/100)*integral(fun4,0,theta/2)+(B/100)*integral(fun5,theta/2,theta);
%Wheel torque(Nm)

fun6 = @(x)( 2.*delay_m.*x.*(((a.*b) ./ sqrt(power(b,2).*power(sin(x),2)...
    +power(a,2).*power(cos(x),2)))./100).*cos(atan(((b^2)*tan(x))/a^2))...
    ./theta-2.*theta_m.*x.*(((a.*b) ./ sqrt(power(b,2).*power(sin(x),2)...
    + power(a,2).*power(cos(x),2))./100).*sin(atan(((b^2)*tan(x))/a^2))./theta ));
fun7 = @(x)( 2.*delay_m.*(theta-x).*(((a.*b) ./ sqrt(power(b,2)...
    .*power(sin(x),2)+power(a,2).*power(cos(x),2)))./100)...
    .*cos(atan(((b^2)*tan(x))/a^2)))./theta- 2.*theta_m.*(theta-x)...
    .*(((a.*b) ./ sqrt(power(b,2).*power(sin(x),2)+power(a,2)...
    .*power(cos(x),2))./100).*sin(atan(((b^2)*tan(x))/a^2)))./theta );
DP = (B/100)*integral(fun6,0,theta/2) + (B/100)*integral(fun7,theta/2,theta);
%Drawbar pull (N)

fprintf('Y =  %f \n',Y);                    %%Vertical wheel load (N)
fprintf('DP =  %f \n',DP);                  %%Drawbar pull (N)
fprintf('I =  %f \n',I);                    %%Wheel torque(Nm)
```

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